

## BEER SHEVA, Israel

WMO: 401900

Lat: 31.2515N

Lon: 34.7995E

Elev: 279

StdP: 98.02

Time Zone: 2.00 (ISR)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 4.8        | 5.8 | -8.1                          | 2.0 | 19.8 | -5.1 | 2.5 | 19.8 | 7.4                   | 11.3 | 6.8 | 12.4 | 1.6                   | 110  | 0.332 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 8             | 11.9                      | 37.5            | 20.0 | 36.0 | 20.6 | 34.8 | 21.0 | 25.1                | 31.0 | 24.3 | 30.2 | 23.7 | 29.5 | 2.9                     | 290  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.7                            | 19.1 | 27.2 | 22.9 | 18.3 | 26.5 | 22.3 | 17.5 | 26.2 | 78.2          | 31.0 | 75.2 | 30.3 | 72.9 | 29.5 | 28.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 6.4               | 5.4  | 4.8 | 3.0                        | 41.4 | 1.4                | 1.1 | 2.0                                                | 42.1 | 1.2        | 42.8 | 0.4        | 43.4 | -0.6       | 44.2 |    |    |
|                   |      |     | 1.4                        | 25.6 | 1.5                | 1.2 | 0.3                                                | 26.5 | -0.6       | 27.2 | -1.4       | 27.9 | -2.5       | 28.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 21.3   | 12.7  | 14.1  | 17.2  | 20.3  | 23.6  | 26.5  | 28.3  | 28.6  | 26.8  | 23.7  | 19.1  | 14.6 |
|                                                                                  | DBStd       | 6.13   | 2.68  | 3.22  | 3.69  | 3.36  | 2.94  | 1.93  | 1.42  | 1.42  | 1.74  | 2.74  | 2.88  | 2.82 |
|                                                                                  | HDD10.0     | 12     | 7     | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2    |
|                                                                                  | HDD18.3     | 529    | 177   | 126   | 67    | 14    | 1     | 0     | 0     | 0     | 0     | 0     | 23    | 122  |
|                                                                                  | CDD10.0     | 4148   | 90    | 116   | 223   | 309   | 422   | 496   | 568   | 577   | 503   | 424   | 274   | 144  |
|                                                                                  | CDD18.3     | 1623   | 1     | 6     | 32    | 73    | 164   | 246   | 309   | 319   | 253   | 167   | 47    | 6    |
|                                                                                  | CDH23.3     | 16529  | 14    | 64    | 359   | 859   | 1708  | 2571  | 3357  | 3507  | 2394  | 1324  | 326   | 44   |
| CDH26.7                                                                          | 7839        | 2      | 13    | 147   | 388   | 806   | 1313  | 1731  | 1763  | 1071  | 517   | 82    | 6     |      |
| Wind                                                                             | WSAvg       | 1.8    | 1.7   | 1.8   | 1.9   | 1.9   | 1.9   | 1.8   | 1.7   | 1.8   | 1.8   | 1.7   | 1.8   | 1.9  |
| Precipitation                                                                    | PrecAvg     | 193    | 49    | 38    | 30    | 10    | 1     | 0     | 0     | 0     | 0     | 6     | 19    | 40   |
|                                                                                  | PrecMax     | 317    | 186   | 115   | 122   | 72    | 10    | 0     | 0     | 0     | 11    | 31    | 82    | 162  |
|                                                                                  | PrecMin     | 64     | 1     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | PrecStd     | 75     | 40    | 28    | 23    | 16    | 3     | 0     | 0     | 0     | 2     | 9     | 18    | 38   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 25.1  | 28.4  | 35.3  | 37.1  | 39.7  | 39.4  | 38.4  | 38.6  | 36.6  | 37.7  | 31.6  | 27.3 |
|                                                                                  |             | MCWB   | 13.8  | 14.5  | 16.2  | 16.5  | 18.6  | 19.0  | 21.3  | 22.3  | 21.2  | 18.7  | 16.4  | 13.1 |
|                                                                                  | 2%          | DB     | 21.8  | 25.2  | 30.7  | 34.2  | 36.2  | 36.3  | 36.2  | 36.1  | 34.5  | 33.5  | 29.0  | 24.3 |
|                                                                                  |             | MCWB   | 11.8  | 13.4  | 15.7  | 15.7  | 18.1  | 19.8  | 21.9  | 23.1  | 22.2  | 19.3  | 16.8  | 12.6 |
|                                                                                  | 5%          | DB     | 19.7  | 22.3  | 27.2  | 31.1  | 33.6  | 34.5  | 35.0  | 34.8  | 33.2  | 31.2  | 27.0  | 21.7 |
|                                                                                  |             | MCWB   | 11.5  | 12.6  | 14.3  | 15.7  | 17.7  | 20.0  | 22.2  | 22.8  | 22.1  | 20.0  | 16.2  | 12.8 |
|                                                                                  | 10%         | DB     | 18.0  | 20.0  | 24.2  | 28.3  | 31.1  | 33.0  | 33.8  | 33.7  | 31.9  | 29.5  | 24.9  | 19.8 |
|                                                                                  |             | MCWB   | 11.2  | 12.1  | 13.8  | 15.0  | 17.5  | 20.0  | 22.0  | 22.6  | 22.0  | 19.6  | 15.4  | 12.4 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 15.3  | 16.6  | 19.0  | 18.7  | 21.3  | 23.7  | 25.5  | 26.5  | 25.5  | 24.0  | 20.7  | 16.3 |
|                                                                                  |             | MCDB   | 19.7  | 21.9  | 28.2  | 26.8  | 30.8  | 31.5  | 31.8  | 33.3  | 31.3  | 29.1  | 25.6  | 20.5 |
|                                                                                  | 2%          | WB     | 13.8  | 14.9  | 16.9  | 17.7  | 20.2  | 22.9  | 24.4  | 25.4  | 24.5  | 22.7  | 19.3  | 15.2 |
|                                                                                  |             | MCDB   | 18.2  | 21.0  | 24.8  | 25.5  | 28.1  | 29.5  | 31.0  | 31.2  | 30.3  | 28.3  | 24.1  | 19.8 |
|                                                                                  | 5%          | WB     | 12.9  | 13.9  | 15.7  | 17.0  | 19.5  | 22.3  | 23.9  | 24.6  | 23.7  | 21.7  | 18.0  | 14.3 |
|                                                                                  |             | MCDB   | 17.1  | 19.5  | 22.9  | 25.6  | 26.9  | 28.7  | 30.1  | 30.2  | 28.9  | 27.4  | 23.0  | 18.4 |
|                                                                                  | 10%         | WB     | 12.1  | 13.1  | 14.8  | 16.3  | 18.9  | 21.7  | 23.3  | 24.0  | 23.1  | 21.0  | 17.0  | 13.4 |
|                                                                                  |             | MCDB   | 16.6  | 17.9  | 21.4  | 24.6  | 26.0  | 28.0  | 29.1  | 29.7  | 28.5  | 26.3  | 22.3  | 17.8 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.0  | 10.7  | 12.0  | 13.6  | 13.5  | 13.1  | 12.2  | 11.9  | 11.9  | 12.3  | 11.5  | 10.6 |
|                                                                                  |             | MCDBR  | 13.4  | 14.8  | 16.8  | 18.7  | 17.6  | 15.9  | 13.7  | 12.9  | 13.2  | 14.2  | 14.0  | 13.4 |
|                                                                                  |             | MCWBR  | 6.1   | 6.3   | 6.6   | 6.4   | 6.0   | 4.6   | 3.8   | 3.3   | 4.1   | 5.0   | 6.1   | 6.1  |
|                                                                                  | 5% WB       | MCDBR  | 10.6  | 11.7  | 13.4  | 14.5  | 13.5  | 13.0  | 12.2  | 11.7  | 11.6  | 11.8  | 11.9  | 10.0 |
|                                                                                  |             | MCWBR  | 5.8   | 5.7   | 6.1   | 5.8   | 4.8   | 3.7   | 3.4   | 2.9   | 3.7   | 4.5   | 5.9   | 5.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.359  | 0.392 | 0.434 | 0.472 | 0.470 | 0.403 | 0.410 | 0.416 | 0.415 | 0.427 | 0.375 | 0.357 |      |
|                                                                                  | taud        | 2.248  | 2.137 | 2.010 | 1.920 | 1.955 | 2.182 | 2.178 | 2.166 | 2.170 | 2.123 | 2.248 | 2.271 |      |
|                                                                                  | Ebn at Noon | 851    | 855   | 844   | 827   | 829   | 882   | 874   | 866   | 852   | 808   | 826   | 832   |      |
|                                                                                  | Edh at Noon | 113    | 139   | 169   | 192   | 187   | 149   | 149   | 148   | 142   | 139   | 112   | 105   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 3.18   | 4.03  | 5.38  | 6.53  | 7.46  | 8.17  | 7.92  | 7.29  | 6.23  | 4.79  | 3.62  | 2.97  |      |
|                                                                                  | RadStd      | 0.17   | 0.24  | 0.28  | 0.22  | 0.28  | 0.10  | 0.10  | 0.10  | 0.15  | 0.17  | 0.17  | 0.15  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (20 neighbors) | +0.45 | N/A     | -2.12  | N/A     | N/A   | N/A   | N/A         | -60     | +151    | +107    |

Nomenclature: See separate page